Problem Statement and Goals
Software Engineering

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Table 1: Revision History

Date	Developer(s)	Change
September 24, 2026	Andreas Sideris	Problem Statement added
September 25, 2026	Andreas Sideris	Inputs and Outputs completed
September 26, 2026	Humna Saeed	Goals and Stretch Goals added
September 26, 2026	Owen Johnson	Stakeholders and Environment completed
September 27, 2026	Humna Saeed	Custom Documents added, Goals edited
September 27, 2026	Andreas Sideris	Reflection added
September 27, 2026	Owen Johnson	Reflection added
September 28, 2026	Owen Johnson	Environment and Stakeholders feedback integrated
September 28, 2026	Humna Saeed	Added reflection
September 28, 2026	Christopher Ludwig	Added reflection
September 28, 2026	Andy	added reflections and team charter changes
November 03, 2026	Owen Johnson	Revised Stakeholders

1 Problem Statement

1.1 Problem

Video games of the roguelite genre have skyrocketed in popularity in recent years, with titles such as *Hades*, *Slay the Spire*, and *Balatro* releasing to widespread

acclaim. This has resulted in the development of many new projects aiming to follow in their footsteps, at times oversaturating the genre with redundant, unengaging gameplay. Due to the unique nature of roguelite video games (users repeat 'runs' of the game after their progress resets from each failed game session, and attempt to upgrade their ability to succeed between each run), poorly designed titles can significantly negatively affect their user experience. This can manifest due to a failure to conduct effective, quantifiable usability testing, to ensure that a game's roguelite features are engaging and satisfying to play. To address this issue, this project will involve the development of a small, unique roguelite video game that provides explicit variation capabilities for its user interface and gameplay, to facilitate more impactful usability testing.

1.2 Inputs and Outputs

Inputs:

1. Users' gameplay control inputs.
2. Usability metric data collected from playtests involving feature variations.
3. Individual feedback collected from playtesters.

Outputs:

1. Using input 1: Gameplay display outputs (user interface, characters and objects, action feedback).
2. Using inputs 2 and 3: Refined gameplay and user interface features.

1.3 Stakeholders

Our stakeholders will fall into 5 distinct categories:

- Capstone course facilitators who give important advice which informs many important project decisions.
- People who have game development experience and can give unique perspectives/feedback.
- People who play video games and are experienced with the roguelike/roguelite genre.
- People who play video games but are not experienced with the roguelike/roguelite genre.
- People who have little to no experience with video games.

These stakeholders will be found both on and off the McMaster campus. Some potential examples of stakeholders include friends, family, classmates, and members of relevant clubs/teams at McMaster.

1.4 Environment

Hardware Environment: The project will be able to run on users' laptops or PCs. The user will interact with the game using either a keyboard + mouse or a game controller with a standard button layout (e.g. XBOX, PlayStation).

Software Environment: The project may be supported on multiple operating systems, such as Windows, MacOS, and various Linux distributions. The environment is composed of the core gameplay loop, which includes interactable characters/objects and various UI elements. The usability data may be collected online through a form.

2 Goals

The main goals of this project include:

- Develop a complete playable roguelite game with a gameplay loop that includes a central hub, enemy encounters, player progression, and the ability to begin a new run after failure.
- Allow players to make strategic decisions during encounters through the use of resources, available actions, and upgrades to the central hub.
- Implement variations in game features across different builds to create different player experiences for usability testing.
- Use usability testing on selected game variations to analyze how different versions affect the player experience.
- Use the results from the usability study to determine which variations improve the overall player experience in a roguelite game.

3 Stretch Goals

The stretch goals of this project include:

- Develop a gameplay simulator that can test variations of game features, enabling comparisons between them and assessing their effects on gameplay diversity and gameplay behaviour.
- Expand the roguelite game by adding additional encounters or mechanics to increase gameplay variety and overall complexity for the player.
- Compare results from simulated gameplay and human gameplay to identify differences and limitations between the two approaches and strengthen conclusions from the conducted usability studies.
- Expand the types of variations being tested, including gameplay, interface features, and control methods, to cover more factors that could affect the player experience and make the usability study more comprehensive.

4 Custom Documents (CDs)

The intended custom documents (CDs) for this project include:

- **Fall**: Feature Oriented Domain Analysis Report
- **Winter**: Usability Report

5 Appendix — Generative AI Use Disclosure

5.1 AI Tools Used

- OpenAI GPT-5.6 Sol
- Google Search AI Overview (Google Gemini)

5.2 How and Where Used

AI tools were used for assistance with LaTeX syntax and commands, during the development of the Problem Statement and Inputs and Outputs sections. AI tools were not used to generate any text or ideas presented in these sections.

5.3 Verification of AI-Generated Content

Any LaTeX syntax provided by AI tools was verified by testing compilation in LaTeX documents, and testing the resulting output. Text was not generated by AI tools, and therefore no AI-generated text was required to be verified.

Appendix — Reflection

Reflection Questions Answered:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

Andreas:

1. While writing the deliverable, our team collaborated very well, with all team members evenly dividing the work to be done, communicating their progress on each task, and completing their work in time. Team members would frequently update the team on their progress on work tasks, and would often collaborate with other members to make larger project decisions in our meetings and Microsoft Teams chat. Our team also regularly communicated with our professor and TAs to clarify any questions we had regarding each part of the deliverable, helping to ensure it was completed to the best of our ability.
2. One pain point I had individually during this deliverable was at times being unsure about the exact requirements for each document section, and experiencing perfectionism when writing my document sections. Initially just completing each section from the information available in the document template, I found that I could also rely on sample documents, the course rubrics, and course lectures to clarify uncertainties on the requirements for each document section, assisting me in completing each section to the best of my ability. I also had issues with being unable to finalize my sections due to a wish to perfect each one, but I was able to resolve this by completing reviews of my own work, and trusting my teammates to find any potential errors in their reviews.
3. Our team adjusted the scope of our goals many times while creating our project idea. Initially planning to just develop a video game, reviewing this concept with Dr. Smith lead us to decide to add further complexity to ensure the project would be suitable for Capstone. This was done by adding the aspect of incorporating explicit source code variations in the game, to facilitate more effective usability testing. The project was also further scoped by all team members discussing what would be possible for the game's features, and deciding on a minimal, possible option that could be extended if extra time is available.

Owen:

1. While writing this deliverable, many things went well. Work was divided efficiently, group members checked in with their progress and checked in on others, and they would ask for help/collaboration when necessary. This led to important decisions being reviewed by more people. While writing, the team was also continuously talking with our contacts, like our Capstone supervisor, course instructor, and TA to ask questions and receive continuous advice/feedback when writing and reviewing the document. This was done both virtually and in-person, which is important since in-person correspondence is a nice resource to have.

2. There were two pain points I can think of for this deliverable. First, I had a slight mental block when asking the group for feedback, since the project is still in its very early stages. I feel like communicating and asking for help when needed is something I need to work on, and I am resolving this by trusting both my group and myself to produce quailty work, which will also get easier as the group works together for longer. The second pain point was with course navigation. This pain point was resolved as I found more resources, such as the rubrics in Avenue to Learn, which were found inside a menu that no other course I've been in has made use of.

3. Our team discussed the scope of the project for a while to ensure the goals are suitable. Using a project proposed from a potential supervisor and modifying it from there was certainly a good start. The team was able to workshop the modified idea with Dr. Welch to ensure he was on board to supervise, while receiving input from Dr. Smith on the feasibility and viability of the project. Feedback from Dr. Smith turned the project idea more towards the goal of usability testing, which we then iterated further with Dr. Welch, with whom we discussed new goals/stretch goals to facilitate this change (like usability testing simulations). Finally, all members of the team played integral roles in scoping the project throughout this process by discussing and providing insight into each other's strengths, weaknesses, and technical proficiencies.

Andy:

1. The process of git (pulling, commiting, pushing) was well maintained throughout the entire process of making the document. The group overall communicated well with eachout and being as transparent as ever. Each member was able to critique and give vital feedback to improve up the written sections.

2. The main painpoint was the Proof of Concept narrowlization. The group needed to explicitly figure out the main risks targets to make sure the PoC is concise as possible. The technologies used needed extensive research and discussion through the team meetings. Meetings as well as information from the supervisors were used as key deciders for narrowing down the concept plan.

3. Due to the new technology of LLMs, extra complexities were added to the scope. Run-time variations as well as other possible features were added to increase complexity.

Humna:

1. This deliverable helped me solidify the project our group aims to pursue. In the earlier brainstorming stages, our group faced doubts and discussions surrounding the project. This deliverable helped scope the project down to its core components, understand the main goals we wish to achieve throughout the capstone course, and identify the scope we hope to pursue, including our usability study. The deliverable helped me understand the bigger "why" behind the project and define it in a clear way. Prior to this deliverable, there was uncertainty about what the project was, how it would come to life, and how we would begin approaching it. By defining the project, it made it feel more tangible and something that can actually be tackled.

2. Our group spent the beginning weeks of capstone solidifying our idea and clearing any existing doubts. Through discussions with our supervisor and group meetings, we developed a clear vision of where the project was headed. Many of these concerns were addressed before completing the deliverable; hence, no major pain points were experienced. For example, we ensured usability studies were included in the project to meet the project requirements, and we finalized key aspects of how the game would work, such as having the central hub play a critical role. It also helped me understand the importance of clear communication moving forward to ensure all members continue to share a unified vision of the project.

3. Our team adjusted the goals to ensure that usability studies played a major role in the project. The scope of our usability study was expanded, and different methods were considered, such as the possibility of utilizing simulated gameplay. The group also decided to implement variations in game features to allow for a more exhaustive range of testing. This helped ensure the project was appropriate for the capstone course based on the project requirements and our discussions with Dr. Smith.

Christopher:

1. While I did not take lead on any section, I did review others' sections and did help rewrite it. I thought we had a very good split for work and we were able to easily communicate our ideas so everyone was able to have their thoughts written on these documents. We were able to compromise very well and in the end we worked very well as a group during meetings and creating the deliverable.

2. A pain point I had was the difficulty in finding certain rubrics and requirements for these documents as there was lots of information spread

out on the avenue page, gitlabs, and lecture slides. Not only the spread out information, but as we reviewed these documents, there were multiple instances of conflicting information, such as one document saying team roles did not need to be assigned and another saying they did.

3. Our scope had to be adjusted multiple times while we were generating ideas and proposing our project as we originally only planned on creating a video game, but we found out this was not enough and had to re-scope to include usability studies to be included in our report. Scope also had to be kept closely in mind when generating ideas for our game as there are many different aspects to game development and we had to limit our ideas to not attempt to incorporate too many features.